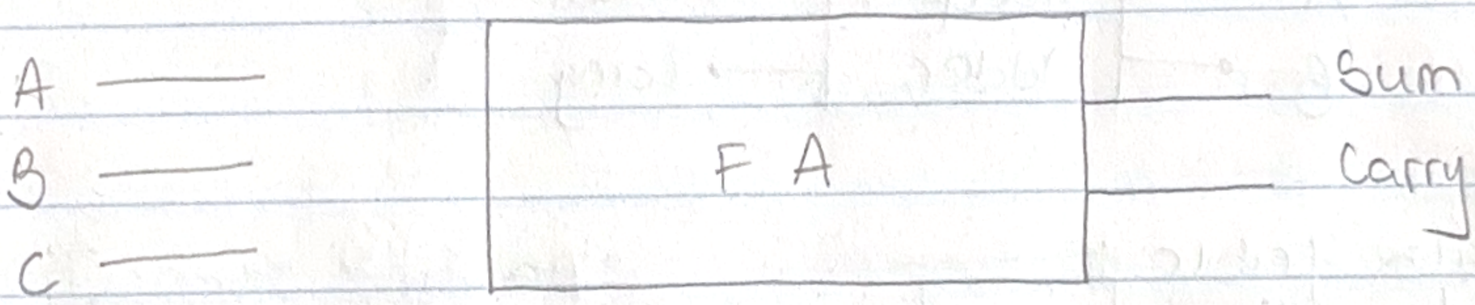
2.) Full Adder: Full adder is a combinational circuit that forms the arithmetic sum of 3 input bits. It consists of 3 inputs and 2 outputs. The third input is Cin which represents the carry from the previous significant position.

Block Diagram:

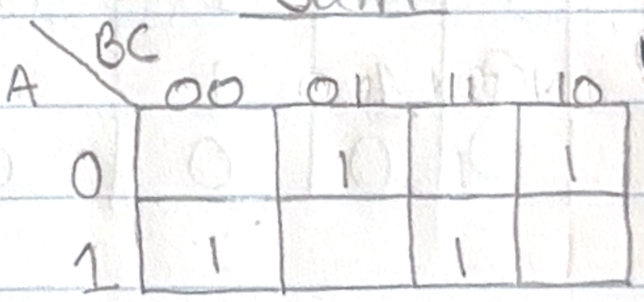


Truth Table

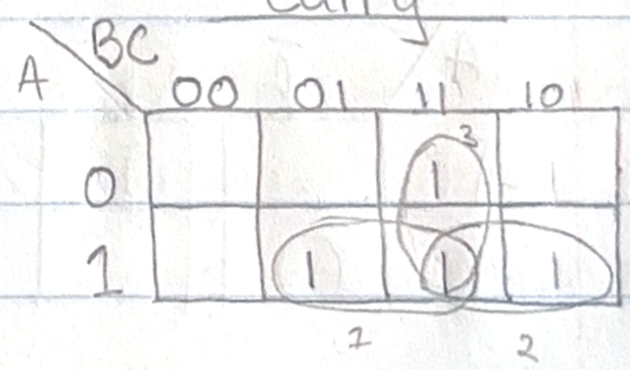
A	B	C	Sum	Carry
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

K Map Simplification

Sum



Carry



a.) $Sum = A\bar{B}\bar{C} + \bar{A}\bar{B}C + \bar{A}B\bar{C} + ABC$

$= A(\bar{B}\bar{C} + BC) + \bar{A}(\bar{B}C + B\bar{C})$ $(\because \bar{X}Y + X\bar{Y} = X \oplus Y)$

$= A\overline{(B \oplus C)} + \bar{A}(B \oplus C)$ $(\because \bar{X}\bar{Y} + XY = \overline{X \oplus Y})$

$= A \oplus B \oplus C$

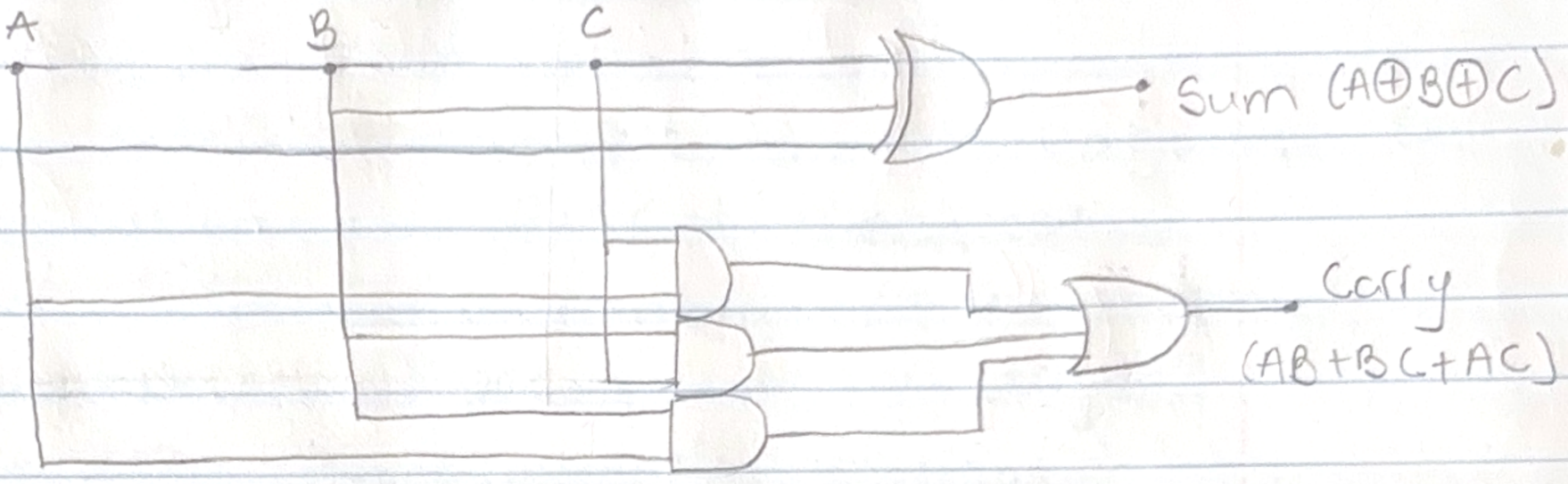
b.) $Carry = (ABC + A\bar{B}C) + (ABC + \bar{A}BC) + (ABC + AB\bar{C})$

Group 1 Group 2 Group 3

* In the Rules of K Map group horizontal and vertical 1's together like so

$Carry = AC + BC + AB$

Logic Circuit



Full Adder circuit

3.) Full Adder using Half Adder :